

Assignment 3: Investment Appraisal

Course: PPIT

Instructions

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- Use the formulas from Lecture 9 to solve the tasks.
- Show all steps clearly.
- Marks are indicated with each part.
- Hand write all solutions, then scan and submit PDF on GCR.

Task 1: Time Value of Money (10 Marks)

You sign a contract to receive Rs. 1,000 after 4 years. The annual discount rate is 8%.

Formula:

$$DF = \frac{1}{(1+r)^t}$$

1. Calculate the discount factor for 4 years. (4 Marks)
2. Find the present value of Rs. 1,000. (6 Marks)

Task 2: Net Present Value (NPV) (15 Marks)

You are offered a contract to buy a van. The project cash flows are:

$$C_0 = -10,000, \quad C_1 = 2,000, \quad C_2 = 3,000, \quad C_3 = 4,000, \quad C_4 = 4,000$$

Discount rate = 10%.

Formula:

$$NPV = \sum_{t=0}^n \frac{C_t}{(1+r)^t}$$

1. Compute the NPV of the project. (10 Marks)
2. State whether the contract should be accepted or rejected based on NPV. (5 Marks)

Task 3: Payback Period (10 Marks)

You are evaluating a software project with the following net cash inflows: Year 1: 1,000; Year 2: 1,500; Year 3: 2,500; Year 4: 3,000. Initial investment = 6,000.

Definition: Payback Period = Time required until cumulative cash flow becomes positive.

1. Compute cumulative cash flows year by year. (5 Marks)
2. Determine the payback period. (5 Marks)

Task 4: Internal Rate of Return (IRR) (15 Marks)

A contract requires an initial investment of Rs. 5,000 and generates Rs. 2,000 annually for 3 years.

Definition: IRR is the discount rate that makes $NPV = 0$.

$$0 = \sum_{t=1}^3 \frac{2000}{(1 + IRR)^t} - 5000$$

1. Set up the NPV equation with IRR. (5 Marks)
2. Solve (approximate) IRR by trial method. (10 Marks)

Task 5: Sensitivity Analysis (10 Marks)

You are evaluating a software project with the following details:

- Initial development cost: Rs. 20,000 (at year 0).
- Selling price per unit: Rs. 5,000.
- Expected sales: Year 1 = 30 units, Year 2 = 35 units, Year 3 = 40 units.
- Discount rate = 10%.

Formula:

$$NPV = \sum_{t=0}^n \frac{C_t}{(1 + r)^t}$$

where C_t = Net Cash Inflows at year t .

Now assume that in Year 3, sales drop from 40 units to 20 units.

1. Recalculate the NPV under this new assumption. (6 Marks)
2. Comment on the project's risk level when sales assumptions change. (4 Marks)

Task 6: Lease vs Buy Decision (15 Marks)

You are considering two contracts for a new car:

- Buy outright for £8,995.
- Lease: £500 down payment plus £400 per month for 24 months.

The monthly discount rate is 0.2466% (equivalent to 3% annual).

Formula:

$$PV = \sum_{t=1}^{24} \frac{400}{(1+r)^t} + 500$$

1. Compute the present value of the lease option. (10 Marks)
2. Decide whether to lease or buy, justifying with the lower cost option. (5 Marks)

Task 7: Contract Writing (25 Marks)

Choose **any one** of the following tasks: **Task 4 (IRR)**, **Task 5 (Sensitivity Analysis)**, or **Task 6 (Lease vs Buy)**.

1. Based on your calculations, write a short contract (in your own words) as if you were preparing an agreement between two parties.
2. The contract should clearly state the terms of investment, expected returns, and decision criteria (e.g., IRR threshold, risk clause, or cost comparison).